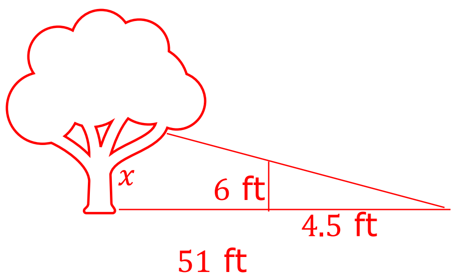
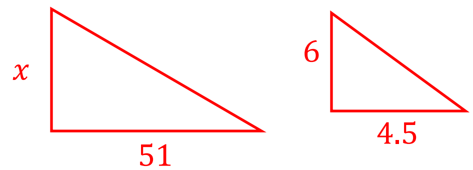
Grade 8 Name Answer Key ___________
Unit 6 Date______________________
Lesson 6 Practice Period_____________________

1. A large tree casts a shadow that is 51 feet long. A 6-foot-tall person standing next to the tree casts a shadow that is 4.5 feet long. How tall is the tree?

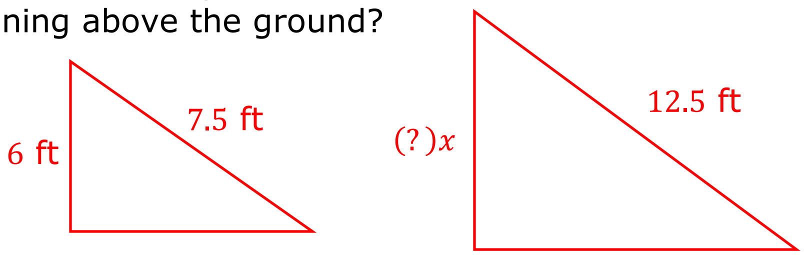




$\frac{x}{6} = \frac{51}{4.5} \rightarrow 4.5x = 51(6) \rightarrow \frac{4.5x}{4.5} = \frac{306}{4.5}$

Tree $= 68$ ft

2. A ladder that is 7.5 feet tall leans against a building at a distance of 6 feet above the ground. If another ladder that is 12.5 feet tall leans against the same building at the same angle what is the distance, in feet, that the taller ladder would be leaning above the ground?

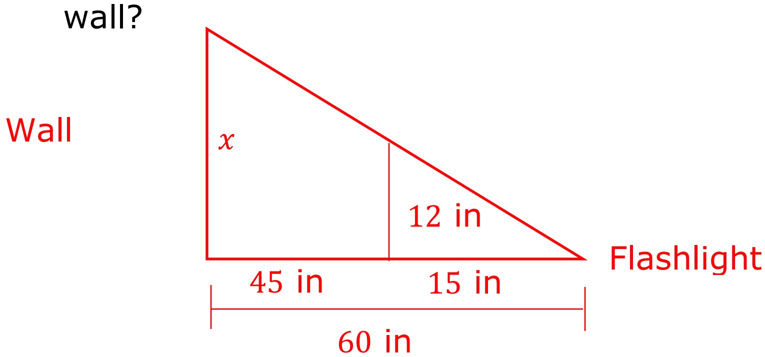


$\frac{6}{x} = \frac{7.5}{12.5}$

$6(12.5) = 7.5x$

$\frac{75}{7.5} = \frac{7.5x}{7.5} = 10$ ft

3. A piece of glass is placed between a flashlight and a piece of paper. The glass is 12 inches tall and is located 45 inches away from the wall and 15 inches away from the flashlight. What is the length of the shadow that would be produced on the wall?



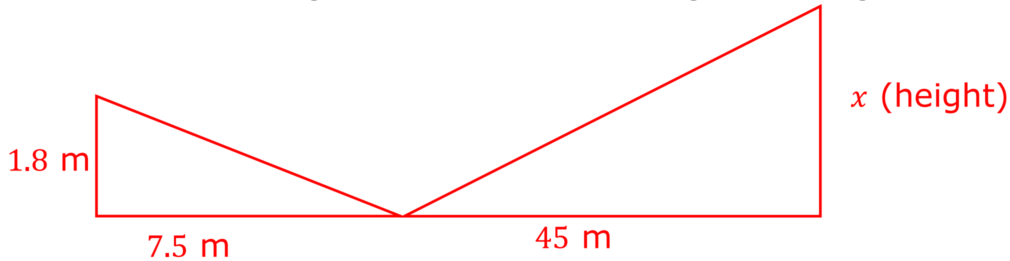
$15\ in + 45\ in = 60\ in$

$\frac{x}{60} = \frac{12}{15}$

$\frac{15x}{15} = \frac{720}{15}$

$x = 48$ inches

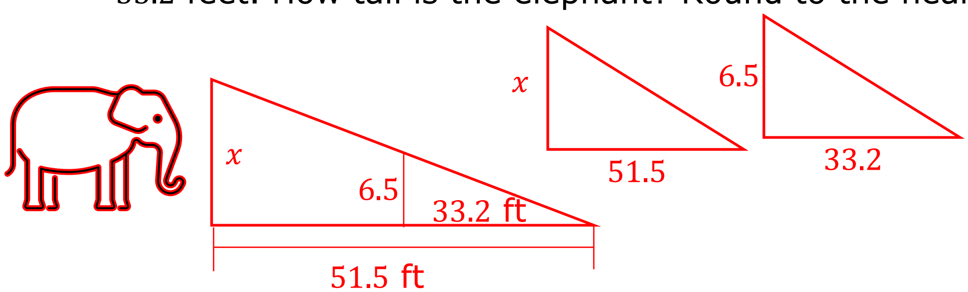
4. A glaciologist is surveying a glacier in Juno, Alaska. She places a glass mirror flat on the ground 45 meters away from the base of the glacier and walks away from the glacier until she is 7.5 meters away from the mirror with her eyes 1.8 meters above the ground. What is the height of the glacier?



$$\frac{1.8}{x} = \frac{7.5}{45}$$
$$1.8(45) = 7.5x$$
$$\frac{81}{7.5} = \frac{7.5x}{7.5}$$

Glacier height $= 10.8$ meters

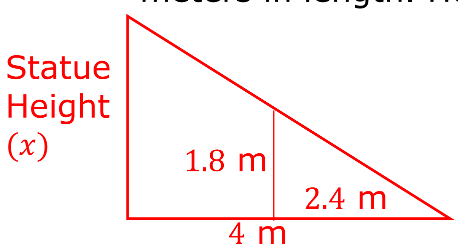
5. During the day, an elephant at the zoo casts a shadow that is 51.5 feet in length. A person taking a photograph near the elephant is 6.5 feet tall and casts a shadow of 33.2 feet. How tall is the elephant? Round to the nearest tenth.



$$\frac{x}{6.5} = \frac{51.5}{33.2}$$
$$33.2x = (6.5)51.5$$

$\frac{33.2x}{33.2} = \frac{334.75}{33.2} = 10.08$ feet

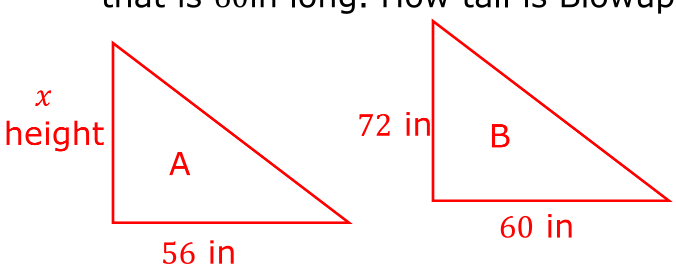
6. A local statue displayed in Town Square casts a shadow that is 4 meters in length. A person who is 1.8 meters tall standing near the statue casts a shadow that is 2.4 meters in length. How tall is the statue?



$$\frac{x}{1.8} = \frac{4}{2.4}$$

$2.4x = 1.8(4) \rightarrow \frac{2.4x}{2.4} = \frac{7.2}{2.4} =$ statue height $= 3$ m

7. Two holiday blowup decorations are placed near each other in a yard. Blowup A casts a shadow that is 56 in long while Blowup B, which is 72 in tall, casts a shadow that is 60in long. How tall is Blowup A?

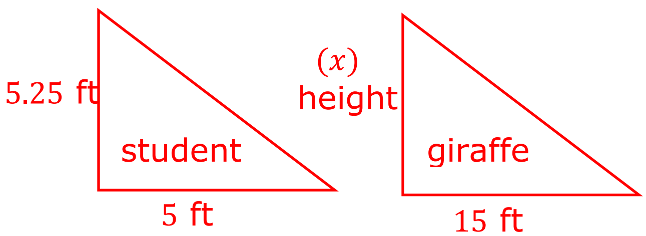


$$\frac{x}{72} = \frac{56}{60}$$
$$60x = 56 \cdot 72$$

$\frac{60x}{60} = \frac{4032}{60} =$ height blowup A $= 67.2$ in

8. A student who is 5 ft 3 in tall notices that her shadow is 5 ft long. At the same time, she notices that the giraffe she is standing near at her local zoo casts a shadow that is 15 ft long. What is the height of the giraffe?

5 ft 3 in $= 5.25$ ft



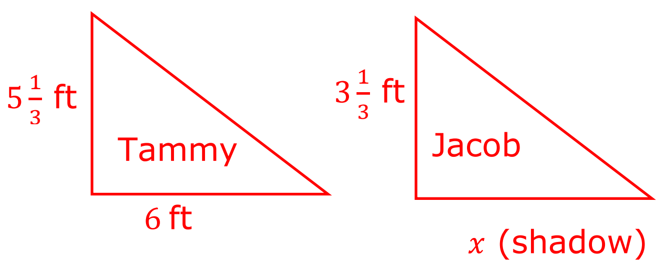
$$\frac{5.25}{x} = \frac{5}{15}$$

$$(5.25)15 = 5x$$

$$\frac{78.75}{5} = \frac{5x}{5} = \text{giraffe height} = 15.75 \text{ ft or}$$

15 ft 9 in

9. Two sisters are outside on a hot day. Tammy, who is 5 ft 4 in tall, casts a shadow that is 6 ft in length. If her sister, Jayne is 3 ft 4 in tall, how long of a shadow will she cast? 5 ft 4 in $= 5\frac{1}{3}$ ft 3 ft 4 in $= 3\frac{1}{3}$ ft



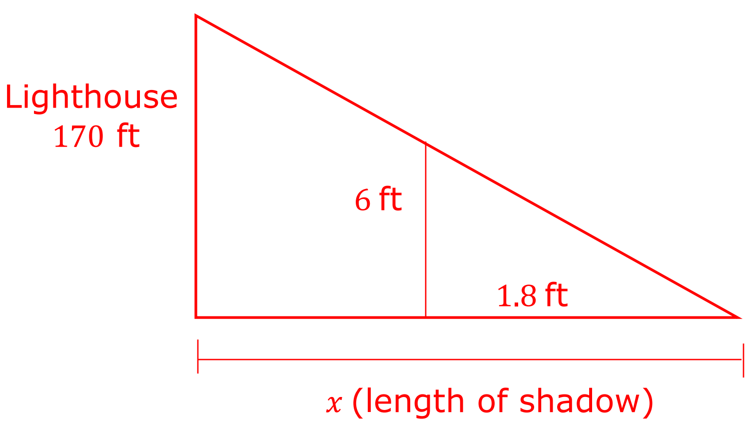
$$\frac{5\frac{1}{3}}{3\frac{1}{3}} = \frac{6}{x}$$

$$5\frac{1}{3}x = 6\left(3\frac{1}{3}\right)$$

$$\frac{5\frac{1}{3}x}{5\frac{1}{3}} = \frac{20}{5\frac{1}{3}}$$

$$x = 3\frac{3}{4} \text{ ft or } 3 \text{ ft } 9 \text{ in}$$

10. A local lighthouse is 170 feet tall. During a visit to the lighthouse Xavier, who is 6 feet tall, cast a shadow that was 1.8 feet long. How long, in feet, was the shadow that the lighthouse cast?



$$\frac{170}{6} = \frac{x}{1.8}$$

$$1.8(170) = 6x$$

$$\frac{306}{6} = \frac{6x}{6}$$

$$x = 51 \text{ ft shadow}$$